

UNIFIED MILITARY ANALYTICS

Power BI Dashboard Storyboard

MODULE 4

Dashboard Planning and Prototyping

A four-page interactive Power BI suite for global military analysis, country profiling, two-country comparison and coalition simulation.

Prepared by

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Data source

military_final.xlsx - 145 countries

Project Overview and Navigation

The dashboard transforms the cleaned and KPI-enriched military dataset into four connected analytical views. Page tabs provide direct navigation, while slicers and disconnected selection tables support dynamic filtering.

Dashboard	Primary objective	Key interaction
Quick Stats	Global overview and rankings	Region, Continent, Alliance and Country filters
Nation Overview	Single-country capability profile	Country dropdown updates all cards and charts
Compare Powers	Side-by-side two-country analysis	Independent Country 1 and Country 2 selectors
Coalition Builder	Aggregate multi-country strength	Multi-select coalition and reference-country selector

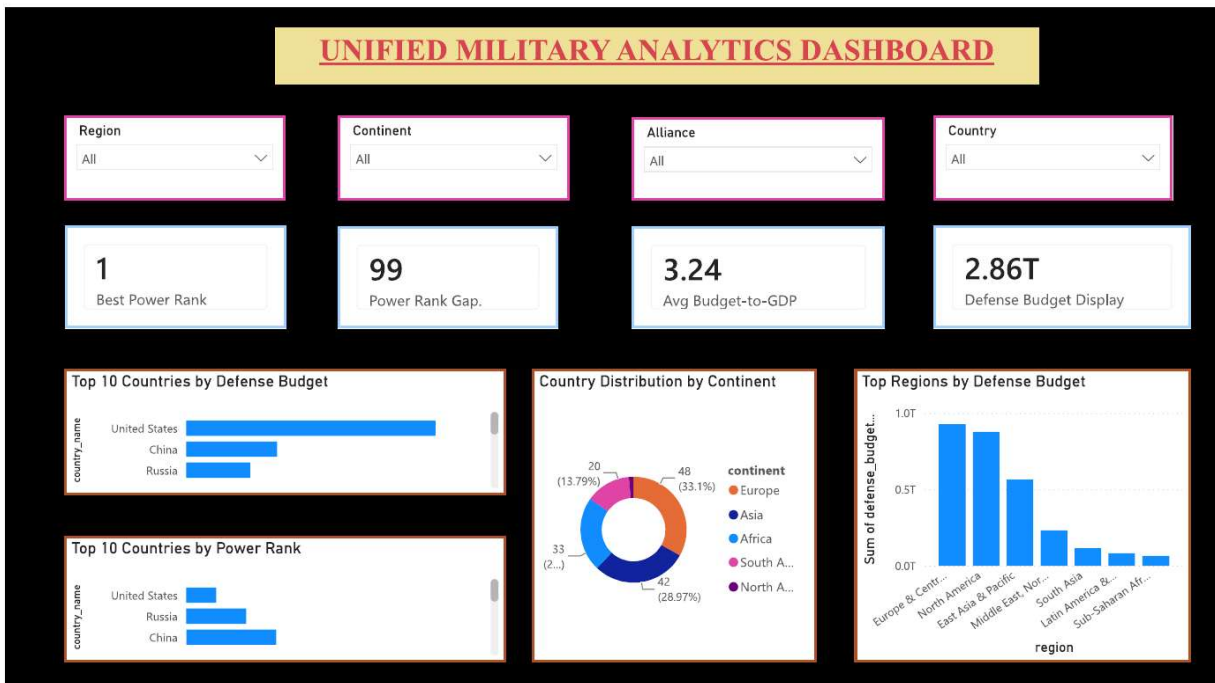
Dashboard Flow

Quick Stats -> Nation Overview -> Compare Powers -> Coalition Builder

Prototype Status

- All four report pages are implemented in Power BI.
- Quick Stats and Nation Overview respond to live slicer selections.
- Compare Powers uses independent selection tables and DAX measures.
- Coalition Builder aggregates selected members and compares them with a reference country.

Dashboard 1 - Quick Stats



PURPOSE

Provides an immediate global overview of military strength, rankings and defense spending.

FILTERS / CONTROLS

- Region
- Continent
- Alliance
- Country

KPI CARDS

- Best Power Rank
- Power Rank Gap
- Average Budget-to-GDP
- Total Defense Budget

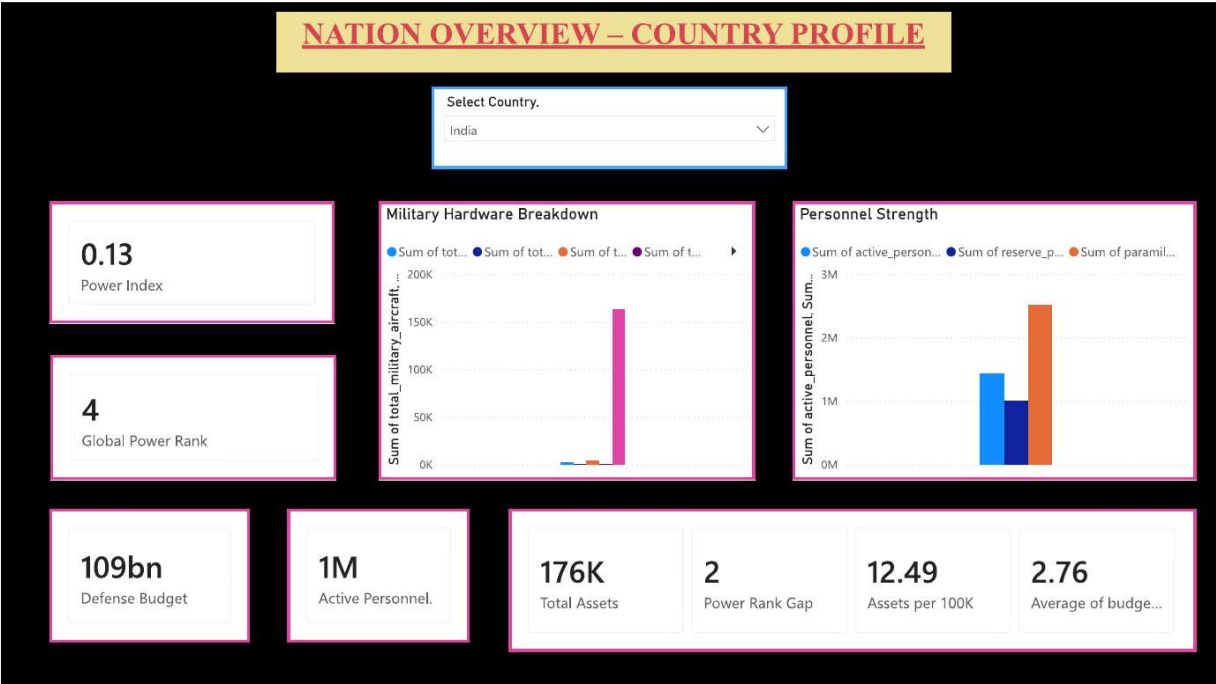
VISUALS

- Top 10 Countries by Defense Budget
- Top Countries by Power Rank
- Country Distribution by Continent
- Top Regions by Defense Budget

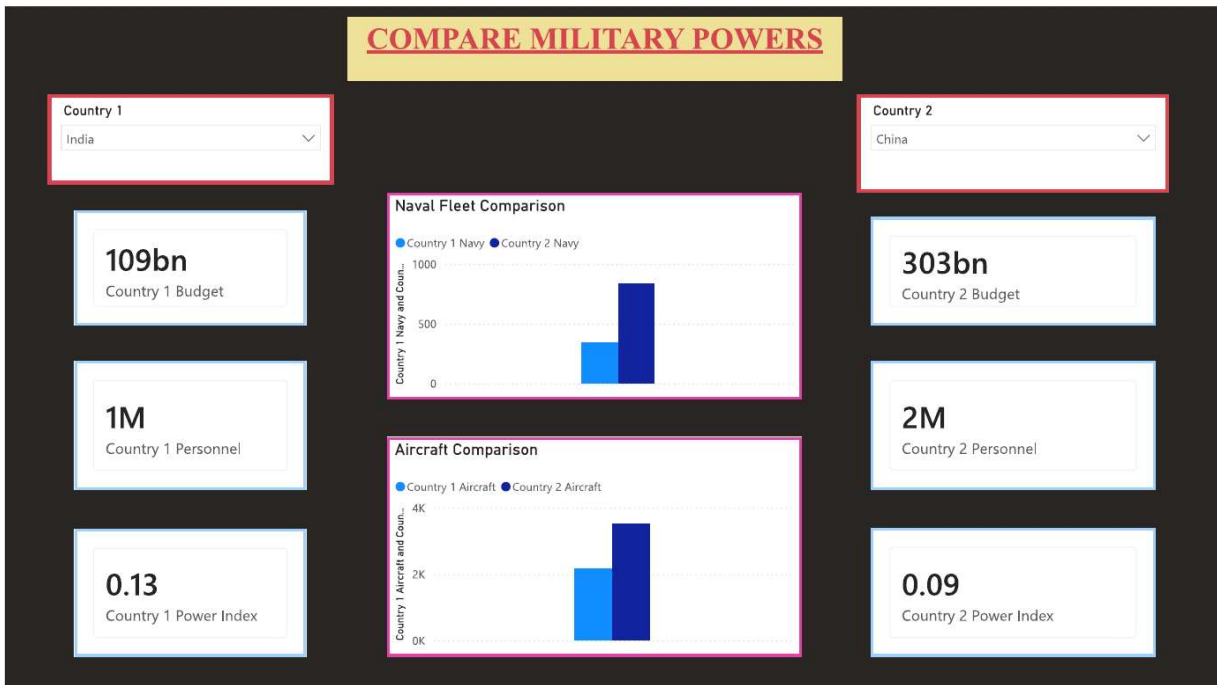
INTERACTION LOGIC

Each slicer filters the KPI cards and all four charts. Selecting a region, continent, alliance or country recalculates the displayed totals and rankings.

Dashboard 2 - Nation Overview



Dashboard 3 - Compare Powers



PURPOSE

Compares two independently selected countries across key military capabilities.

FILTERS / CONTROLS

- Country 1 dropdown
- Country 2 dropdown

KPI CARDS

- Country 1 and Country 2 Defense Budget
- Country 1 and Country 2 Personnel
- Country 1 and Country 2 Power Index

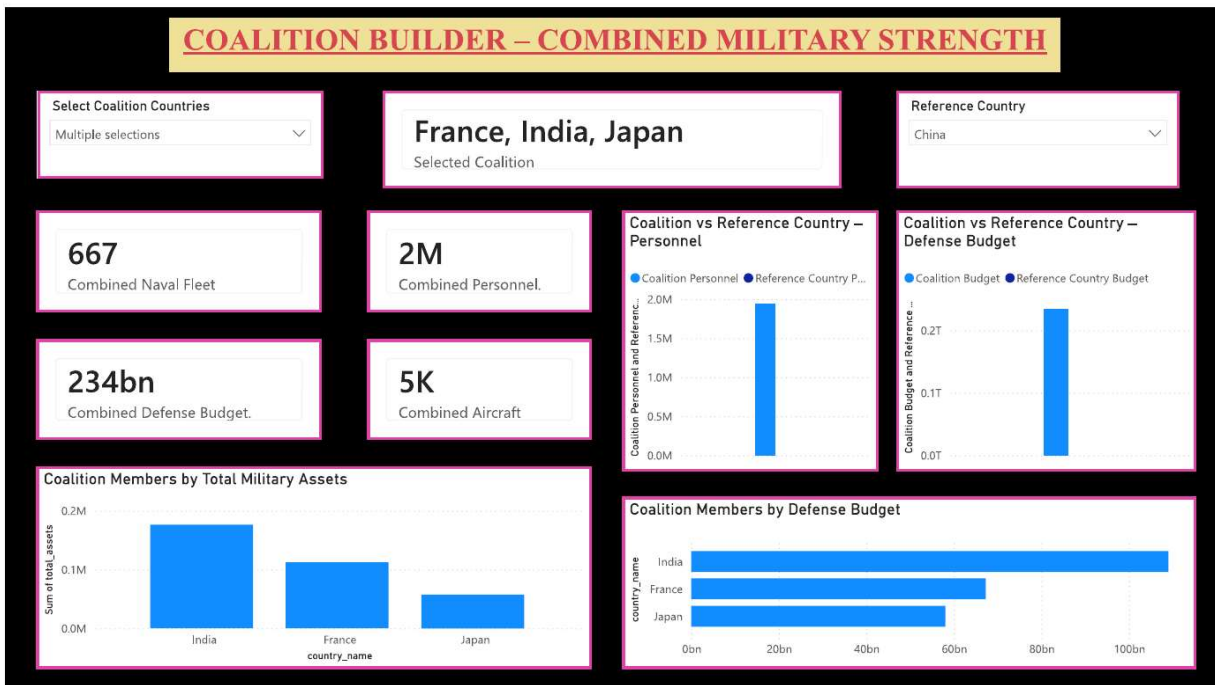
VISUALS

- Naval Fleet Comparison
- Aircraft Comparison

INTERACTION LOGIC

Disconnected Country 1 and Country 2 tables drive DAX measures through TREATAS, allowing both selections to operate independently.

Dashboard 4 - Coalition Builder



PURPOSE

Combines multiple countries into a coalition and evaluates their aggregated strength.

FILTERS / CONTROLS

- Multi-country Coalition selector
- Single-select Reference Country

KPI CARDS

- Selected Coalition members
- Combined Defense Budget
- Combined Personnel
- Combined Aircraft and Naval Fleet

VISUALS

- Coalition Members by Total Military Assets
- Coalition Members by Defense Budget
- Coalition vs Reference Country - Personnel
- Coalition vs Reference Country - Defense Budget

INTERACTION LOGIC

Adding or removing a coalition member recalculates all totals and member charts. The reference selector provides an independent benchmark for comparison.

Final Prototype Summary

Requirement Coverage

- Storyboard covers Quick Stats, Nation Overview, Compare Powers and Coalition Builder.
- Filters, KPI cards, charts and navigation logic are mapped.
- The prototype uses the Module 3 Wide_Format dataset.
- Operational interactions are implemented across all four pages.
- Dashboard values were spot-checked against military_final.xlsx.

Key DAX Logic

- Defense Budget Display formats the global budget total in trillions.
- Disconnected country tables enable independent two-country comparison.
- TREATAS maps selected comparison countries to the main Wide_Format table.
- CONCATENATEX displays selected coalition-member names.
- Coalition cards aggregate the currently selected member countries.

Deliverables

- military_dashboard_prototype.pbix - interactive Power BI dashboard
- military_final.xlsx - processed analytical dataset
- Storyboard.pdf - this dashboard planning document
- README.md - setup and usage documentation

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Project	Unified Military Analytics
Module	Module 4 - Dashboard Planning and Prototyping